

0/1 Knapsack - Dynamic Programming

The 0/1 Knapsack problem is a classic optimization challenge. We are given n items, where each item i has a weight w_i and a value v_i . We also have a Knapsack with a maximum weight capacity W .

Our goal is to select a subset of these items to put into the Knapsack such that the total weight does not exceed W , and the total value is maximized. The "0/1" means we cannot break items into fractions - we either take an item (1) or leave it (0).

To solve this optimally, we use Dynamic Programming. Let's define our state:

$dp[i][j]$ = the maximum value achievable using the first i items and a knapsack capacity of j .

For each item, we face a simple choice: include it or exclude it. This gives us our recurrence relation. If the current item's weight w_i is less than or equal to the current capacity j , we consider both options. If it's too heavy, we are forced to exclude it.

$$dp[i][j] = \max(dp[i-1][j], dp[i-1][j-w_i] + v_i)$$

Let's walk through a small example. We have 4 items and a Knapsack capacity of $W = 8$.

Item 1: $w_1 = 2, v_1 = 3$

Item 2: $w_2 = 3, v_2 = 4$

Item 3: $w_3 = 4, v_3 = 5$

Item 4: $w_4 = 5, v_4 = 6$

We will build a DP table where rows represent the items (0 to 4) and columns represent the capacity (0 to 8). The 0th row and 0th column are initialized to 0 because a knapsack with 0 capacity holds 0 value, and 0 items provide 0 value.

	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	1	0	0	3	3	3	3	3	3
2	2	0	0	3	4	7	7	7	7
3	3	0	0	3	4	5	8	9	9
4	4	0	0	3	4	5	8	9	10

Let's trace the computation for the cell $dp[3][6]$ (which corresponds to considering Item 3 with a capacity of 6).

Item 3 has $w_3 = 4$ and $v_3 = 5$. Our current capacity $j = 6$.

Choice 1 (Exclude Item 3): We simply carry over the best value from the previous row for the same capacity.
 $dp[2][6] = 7$

Choice 2 (Include Item 3): We add the value of Item 3 (5), but we must deduct its weight (4) from our current capacity to see what was the best value we could get with the remaining capacity.

$$dp[2][6 - 4] + 5 = dp[2][2] + 5 = 3 + 5 = 8$$

We take the maximum of these two choices: $\max(7, 8) = 8$.
Therefore, $dp[3][6] = 8$.

The final answer to our problem is found in the bottom-right cell of the table, $dp[4][8]$, which is 10.

At every step, we ask: "Is it better to include this item or not?" We don't